Machine Learning-5

SVM = Support Vector Machine

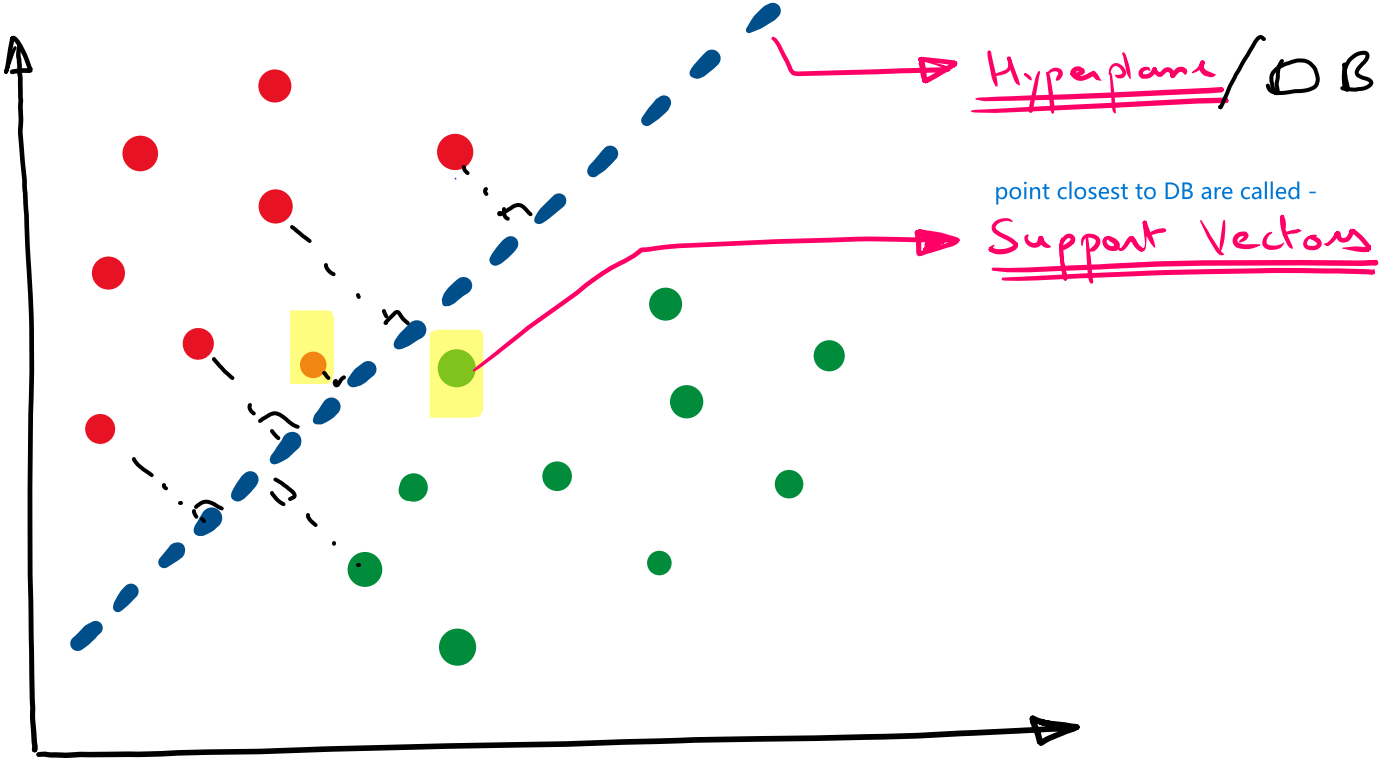
SVM is used where-

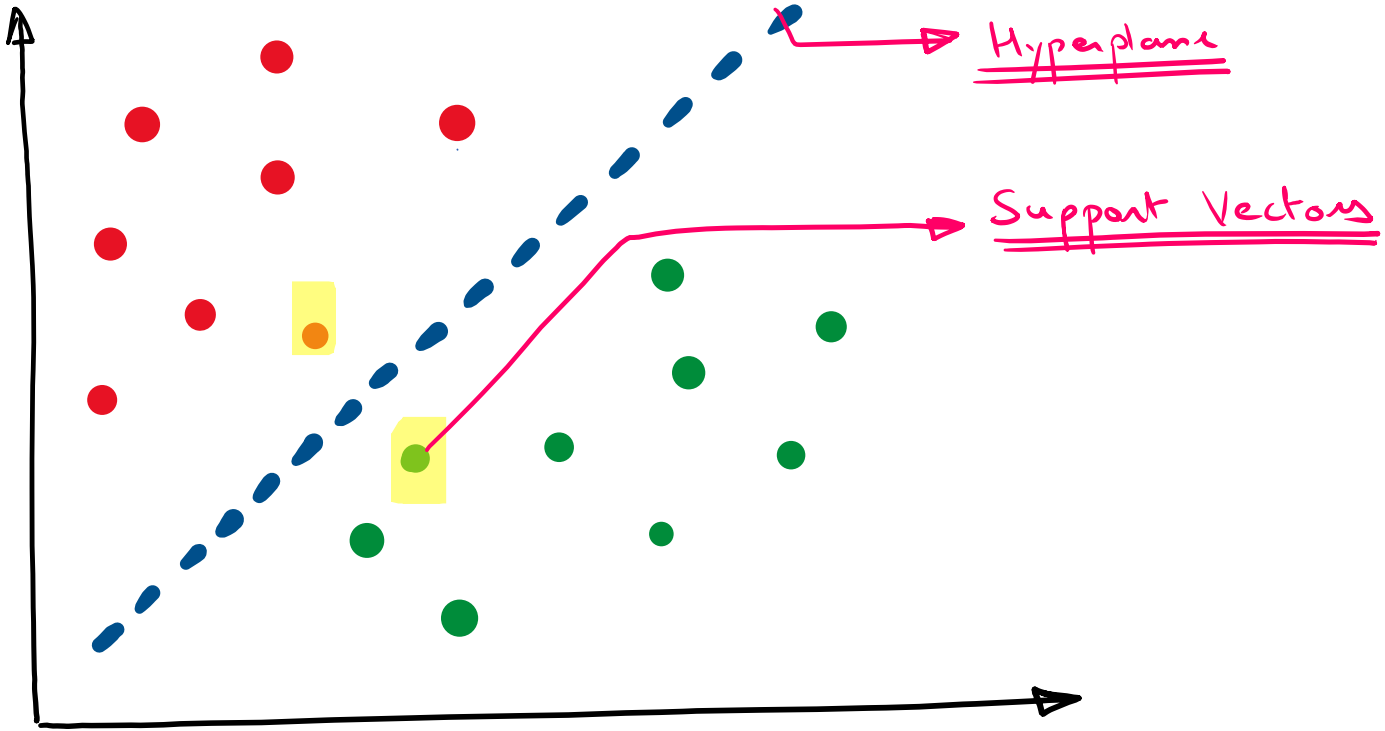
Number of variables > Number of rows

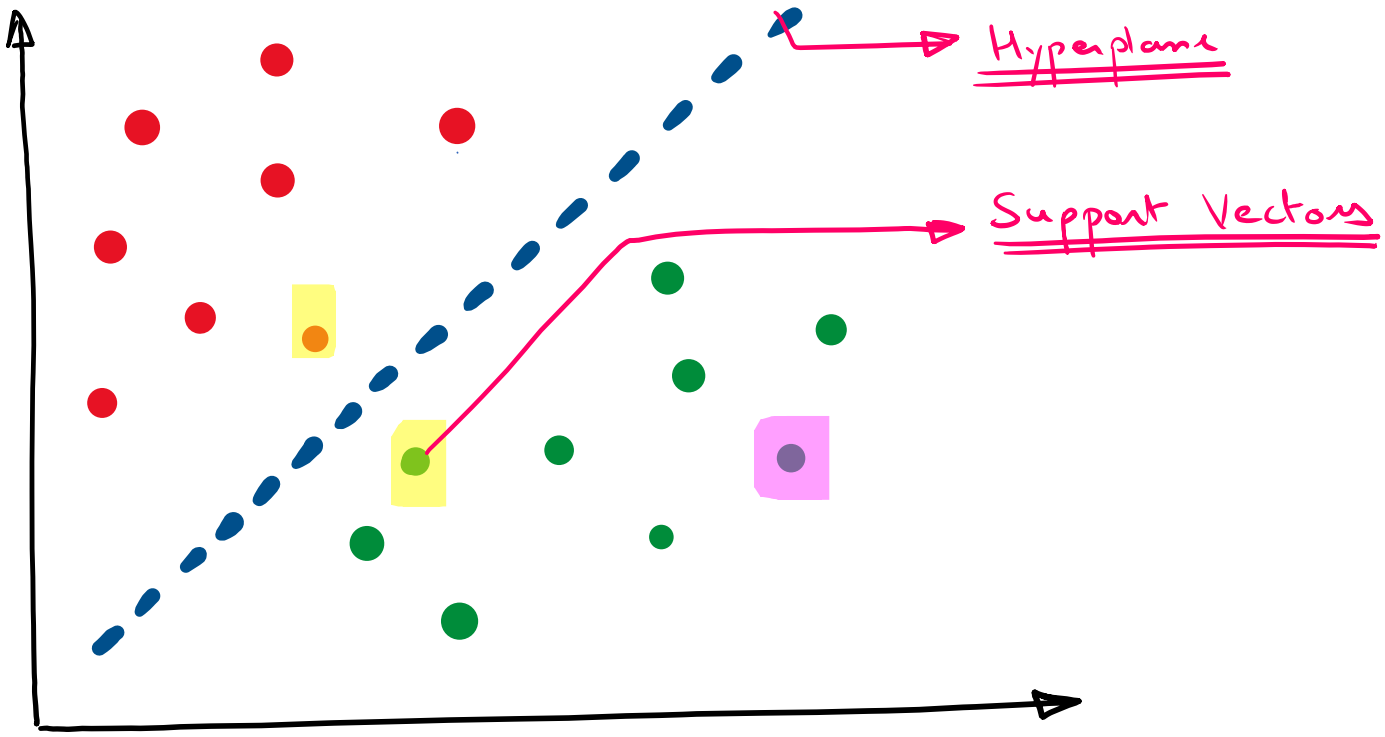
#col > #Rows

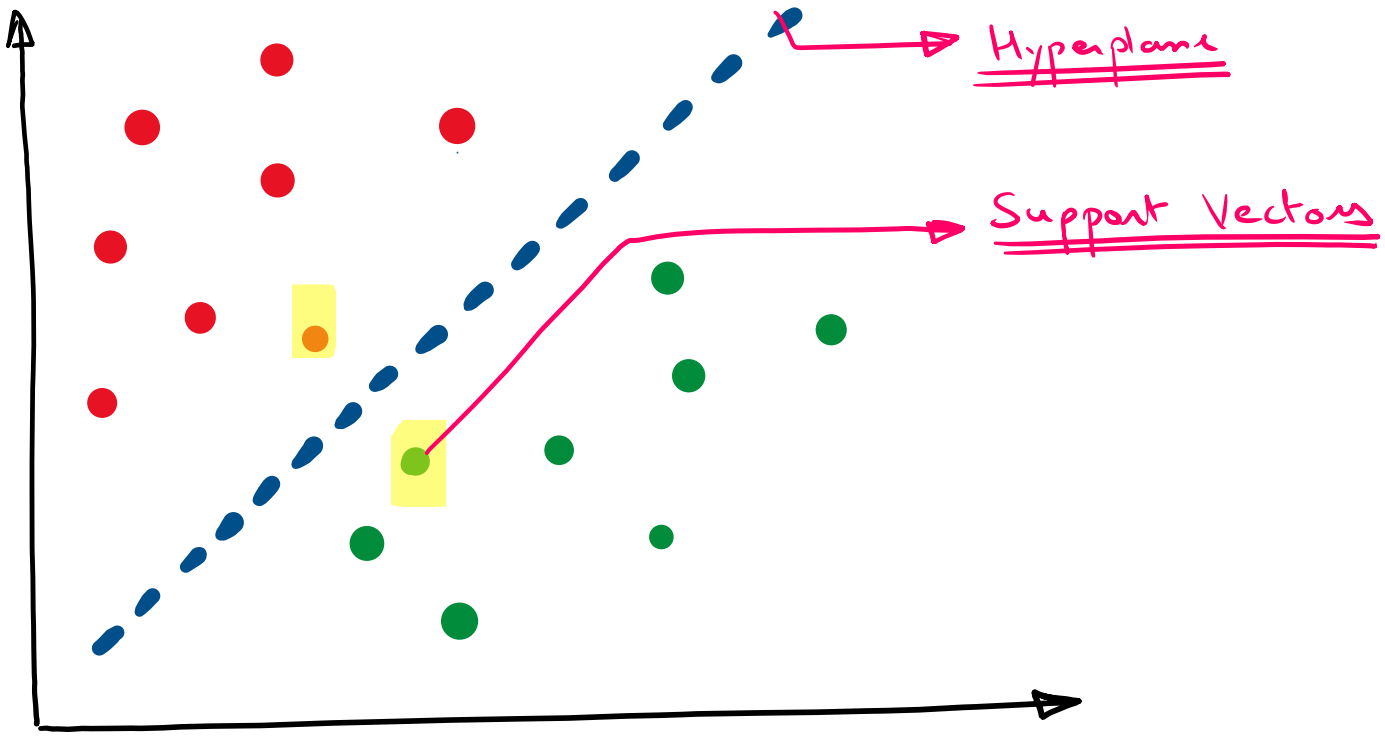
incase this happens SVMs outperform other ML tech

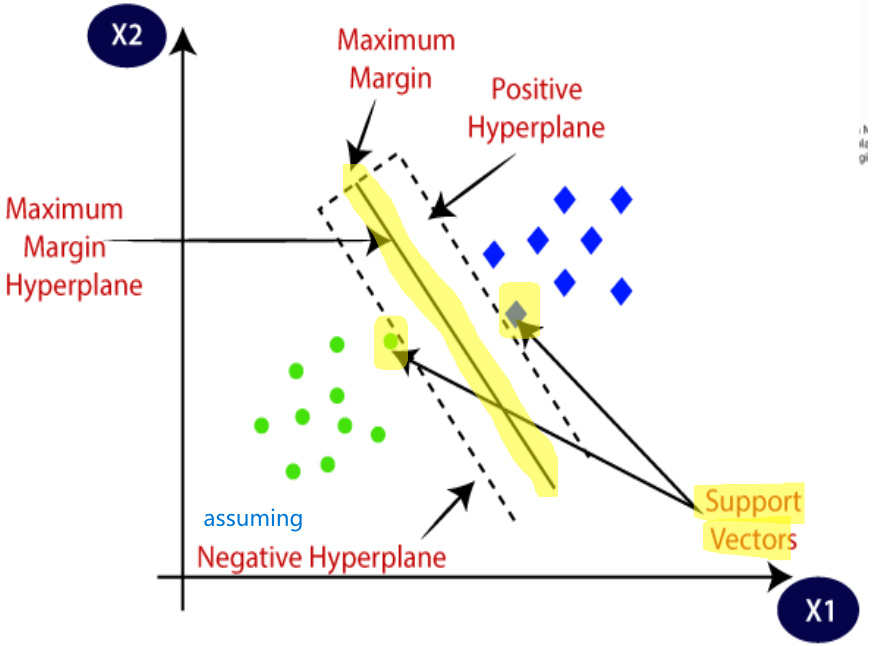
terms


Hyperplane/DB
point closest to DB are called -
Support Vectors


Hyperplane
Support Vectors


Hyperplane
Support Vectors

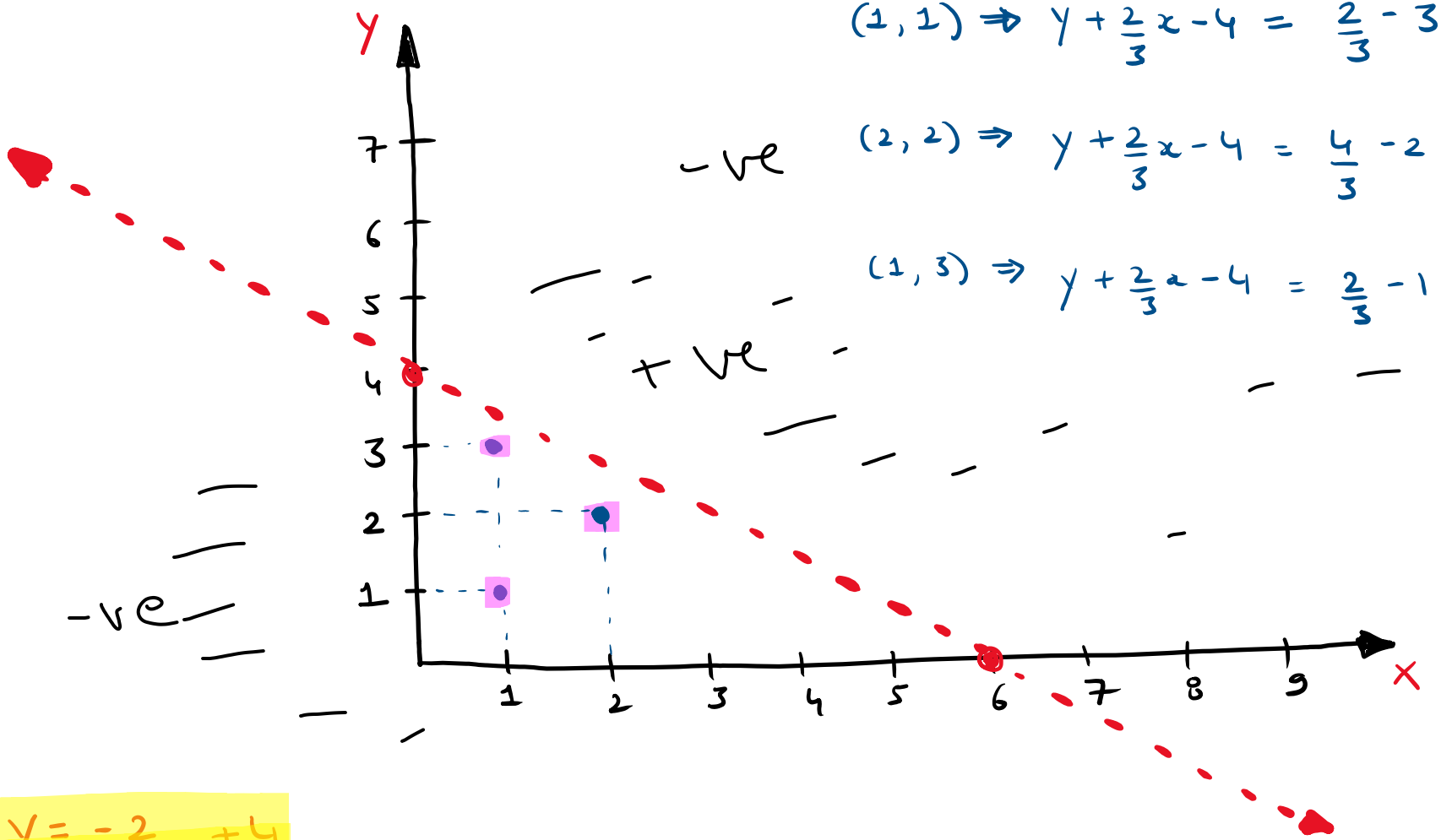

Hyperplane
Support Vectors


X2
Maximum
Margin
Positive
Hyperplane
Maximum
Margin
Hyperplane
assuming
Negative Hyperplane
Support
Vectors
X1

$(1,1) \Rightarrow y + \frac{2}{3}x - 4 = \frac{2}{3} - 3 < 0$

$(2,2) \Rightarrow y + \frac{2}{3}x - 4 = \frac{4}{3} - 2 < 0$

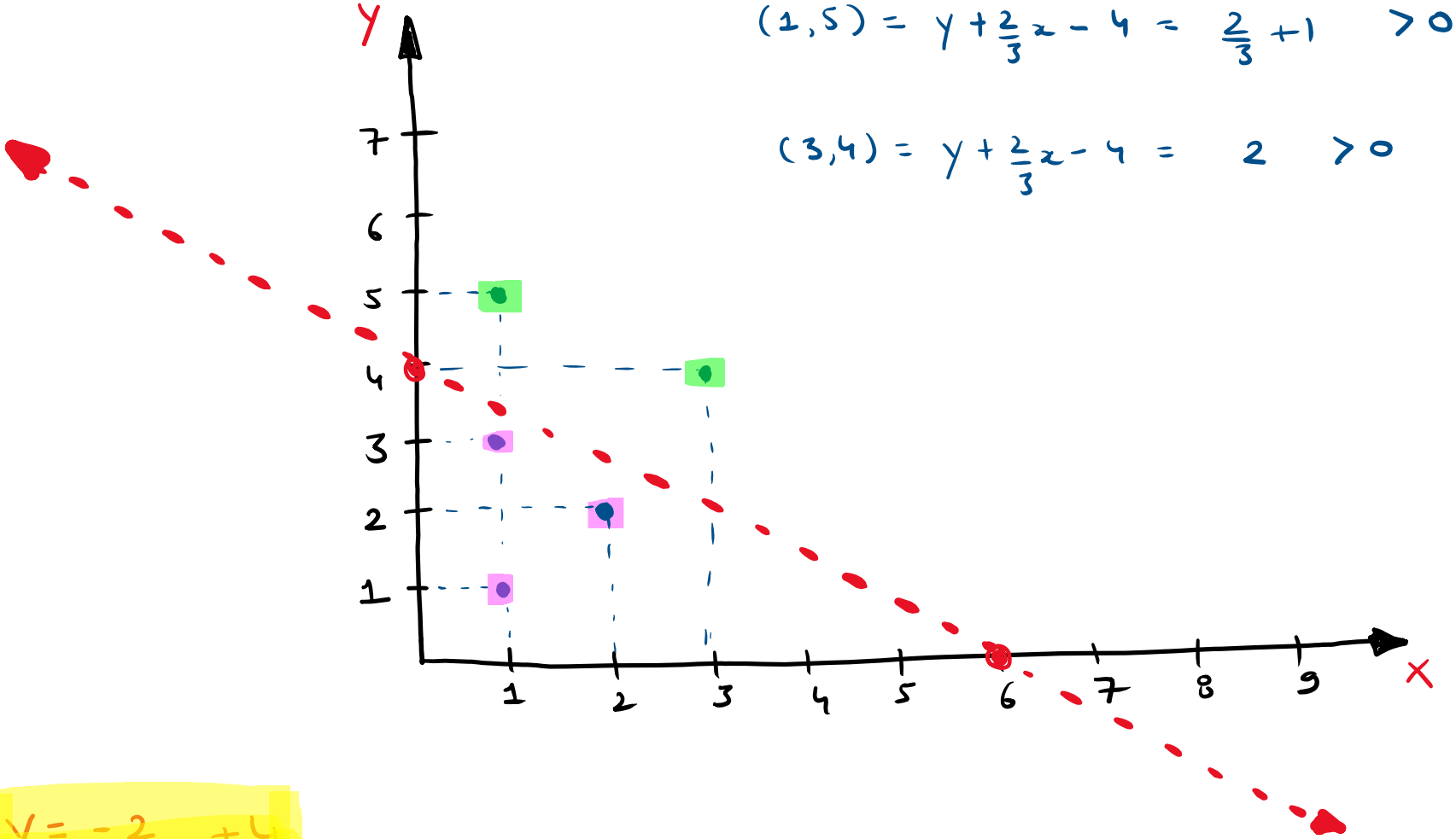
$(1,3) \Rightarrow y + \frac{2}{3}x - 4 = \frac{2}{3} - 1 < 0$



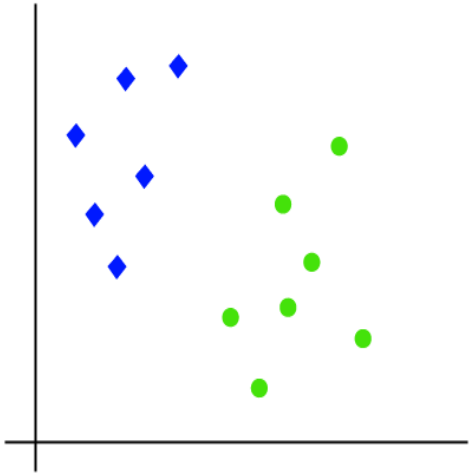
$y = -\frac{2}{3}x + 4$

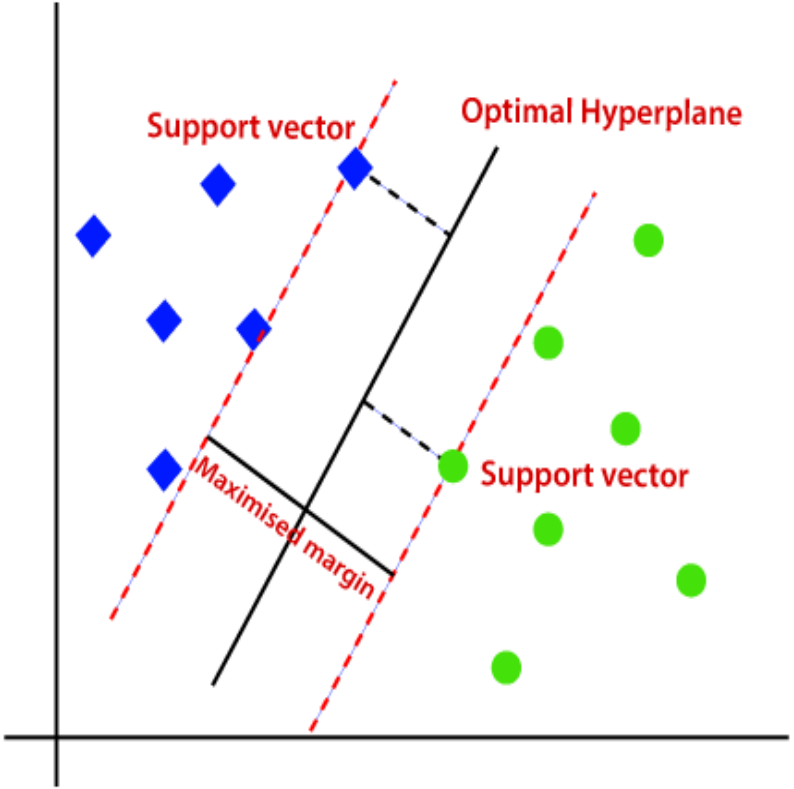
$$(1,5) = y + \frac{2}{3}x - 4 = \frac{2}{3} + 1 > 0$$

$$(3,4) = y + \frac{2}{3}x - 4 = 2 > 0$$

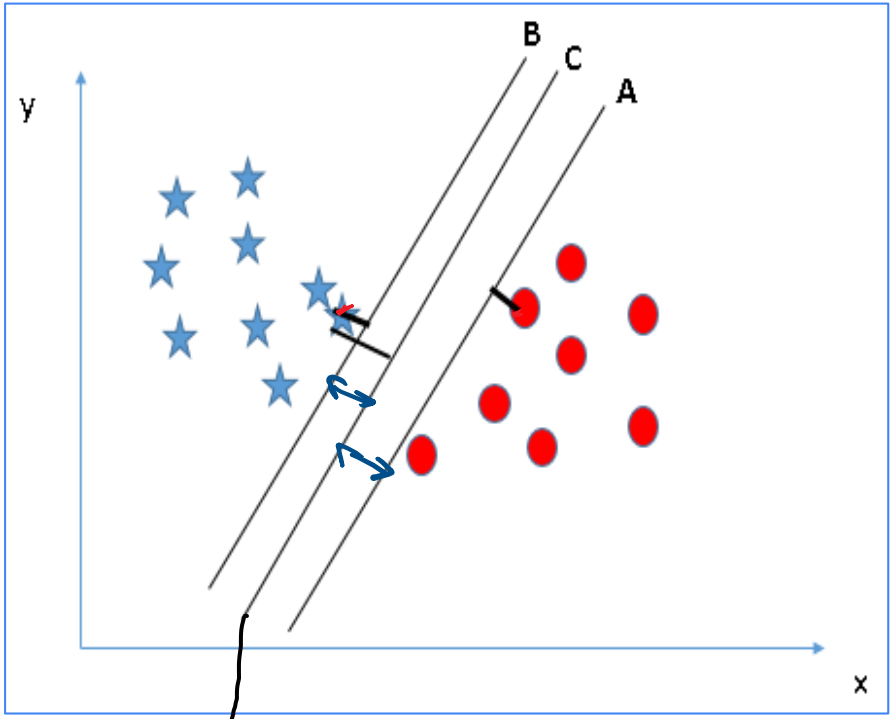


$$y = -\frac{2}{3}x + 4$$

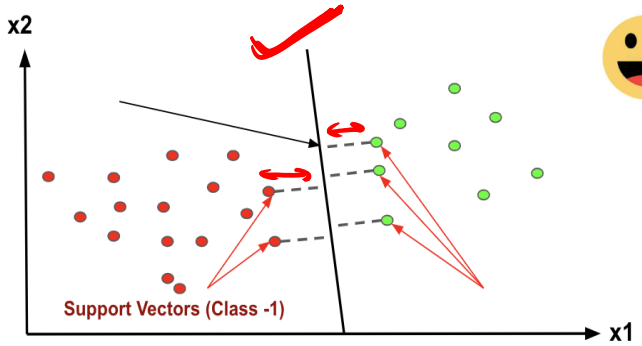



Support vector
Optimal Hyperplane
Support vector
Maximised margin

How to select a hyperplane?

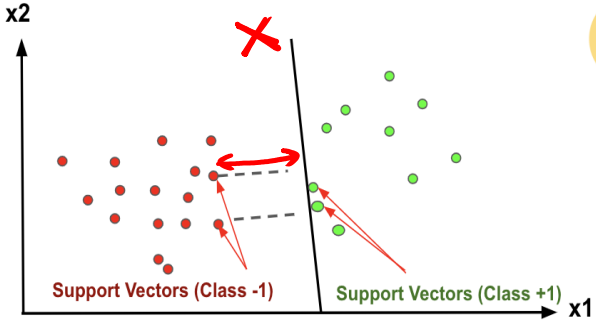


Best QB



Good Margin

- all sector vectors have the same distance with the maximum margin hyperplane

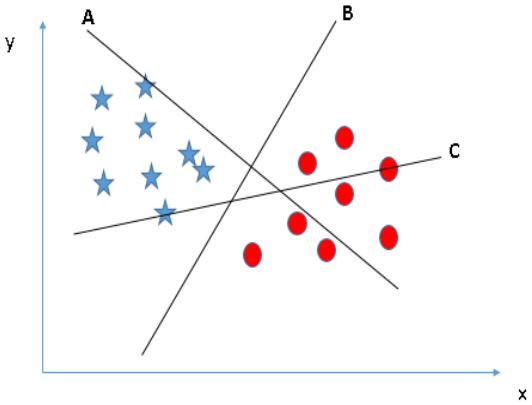


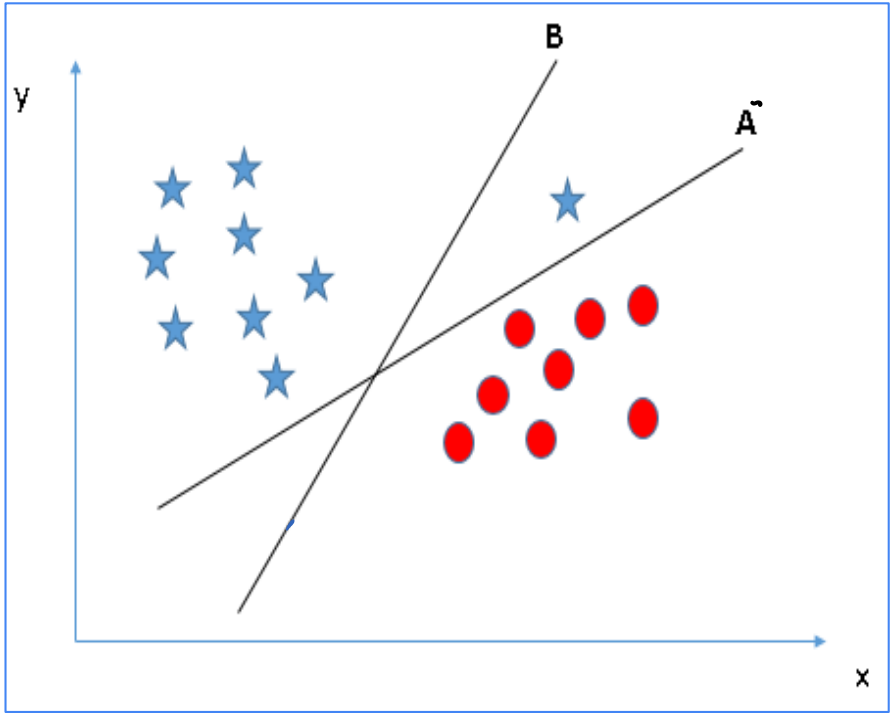
Bad Margin

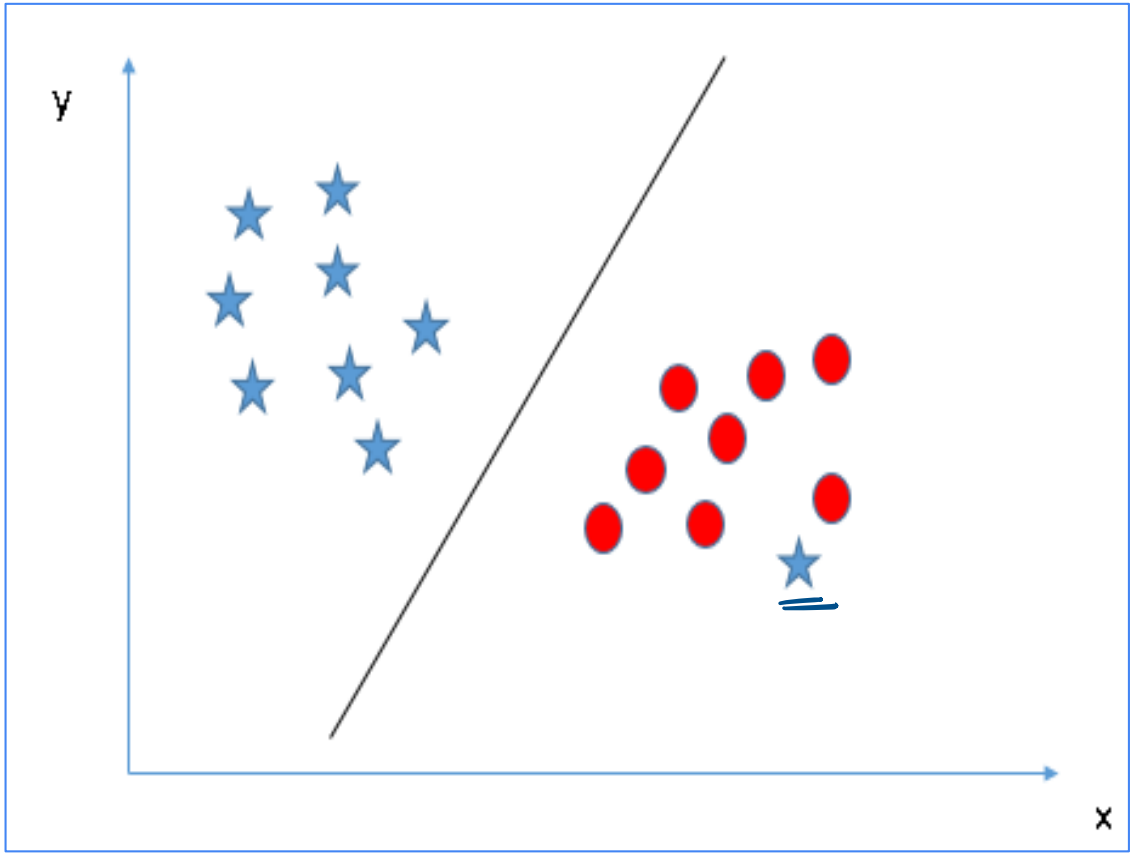
- very close to either class -1 support vectors or class +1 support vectors

How to select a hyperplane?

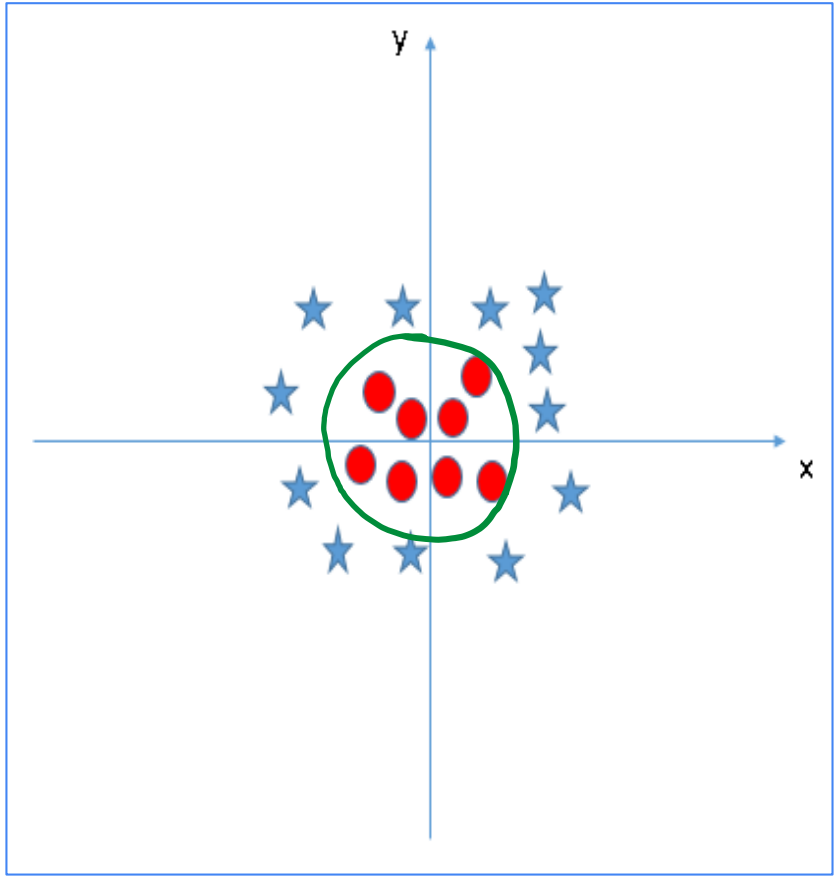
Best D B?

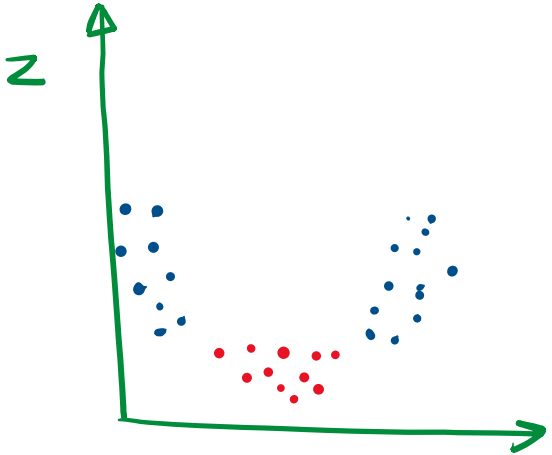


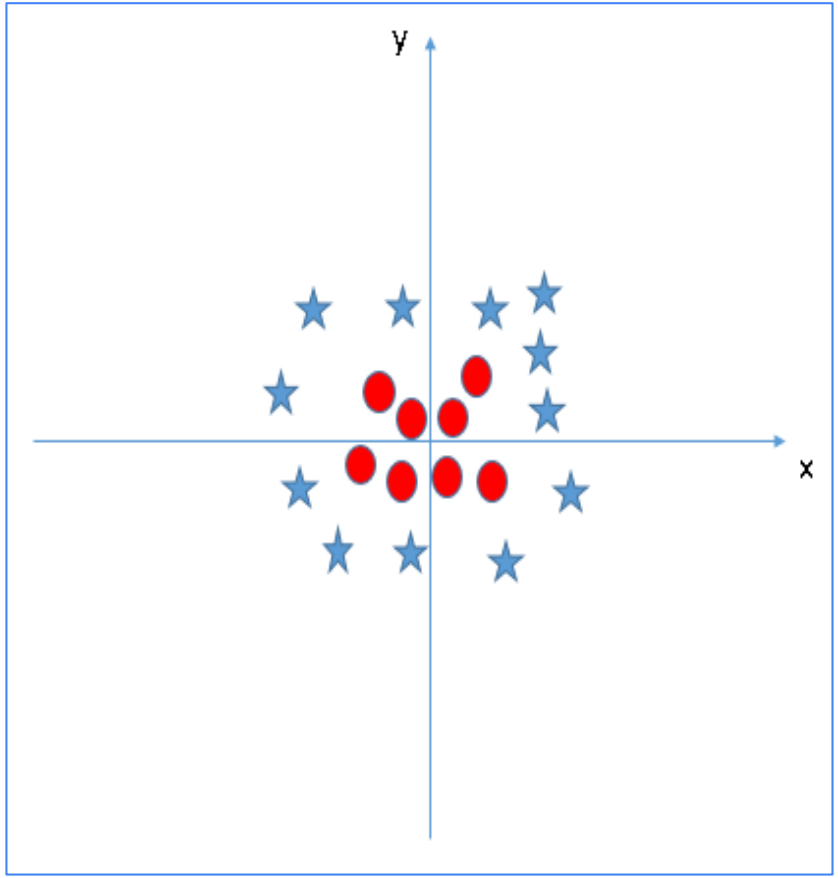

B
A
y
x

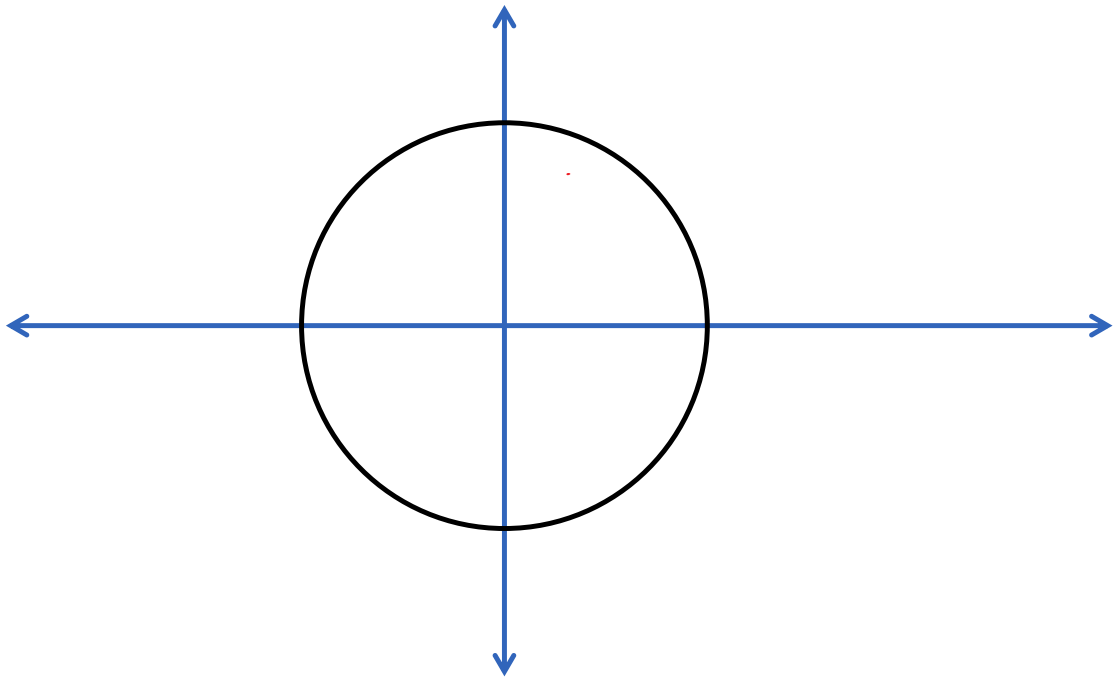

y
x

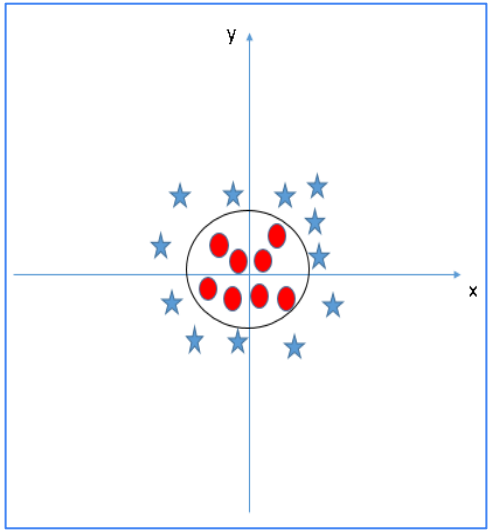
new dimension $Z = x^2 + y^2$



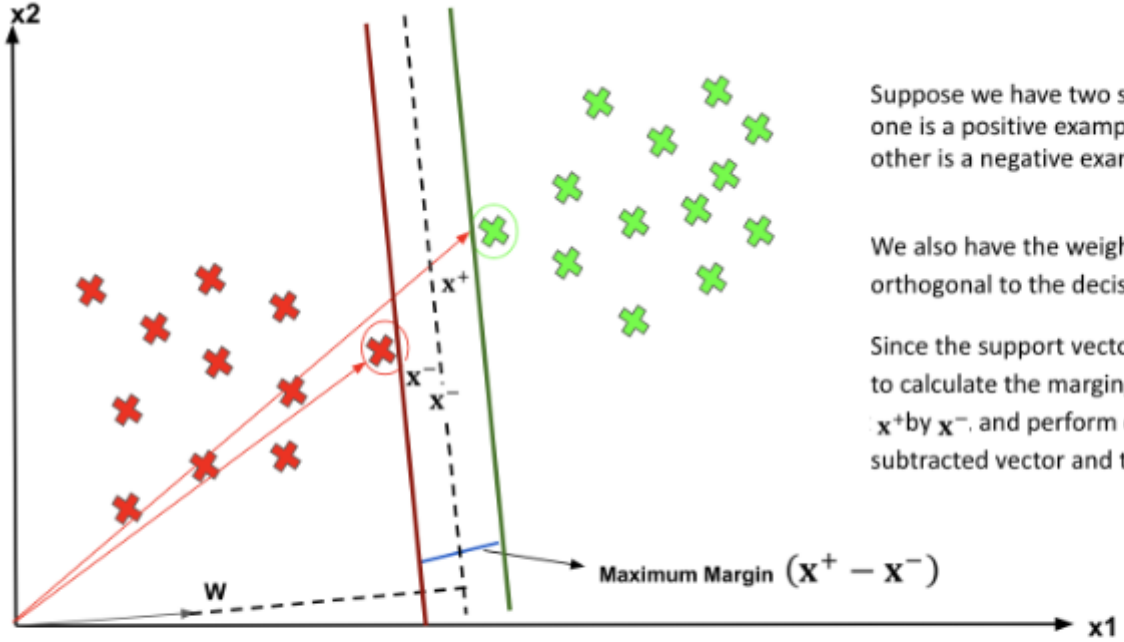



y
x




y
x

inf number line are possible with 100% acc I want link with max margin



Suppose we have two support vectors where one is a positive example (e.g., $\mathbf{x}^+$) and the other is a negative example (e.g., $\mathbf{x}^-$)

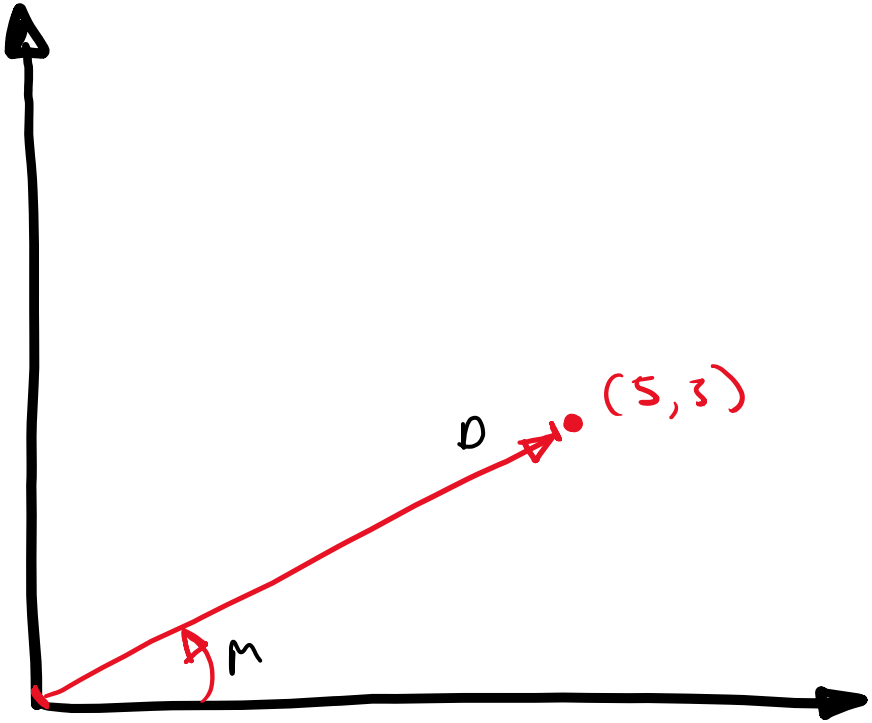
We also have the weight vector $\mathbf{w}$ which is orthogonal to the decision boundary

Since the support vectors define the margin, to calculate the margin, we can first subtract $\mathbf{x}^+$by $\mathbf{x}^-$. and perform dot product on the subtracted vector and the unit vector of $\mathbf{w}$

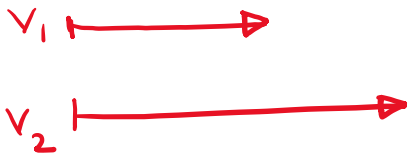
$$(\mathbf{x}^+ - \mathbf{x}^-) \cdot \widehat{\mathbf{w}} = (\mathbf{x}^+ - \mathbf{x}^-) \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} = \mathbf{x}^+ \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} - \mathbf{x}^- \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|}$$

Support Vector Machine

Vector = Magnitude & Direction



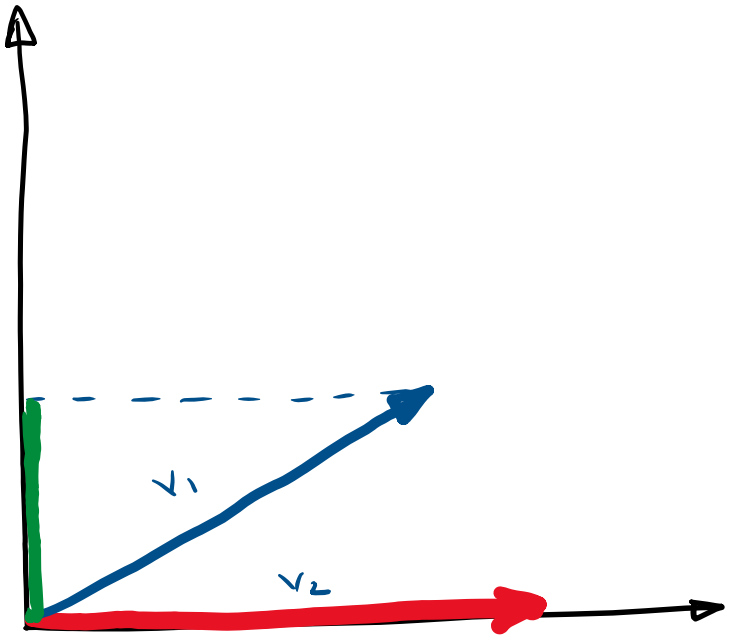
$$\sqrt{5^2 + 3^2}$$


v_1
v_2

$v_1 + v_2$

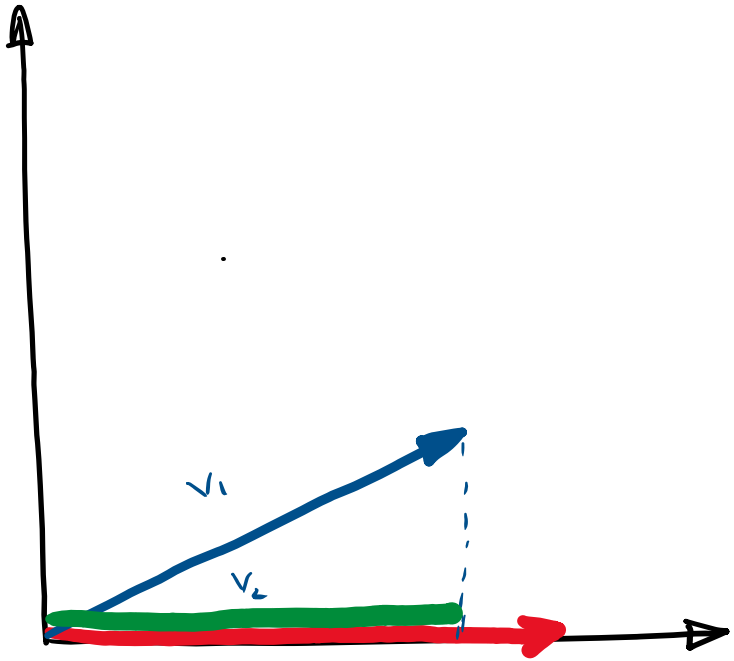
$v_1 - v_2$
=

CROSS PRODUCT



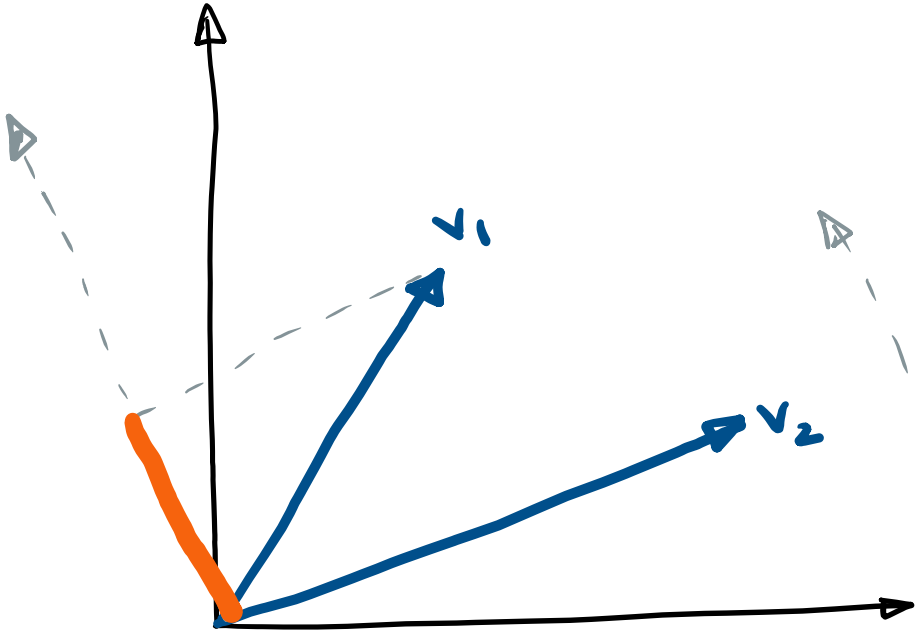
$$v_1 \times v_2$$

DOT PRODUCT



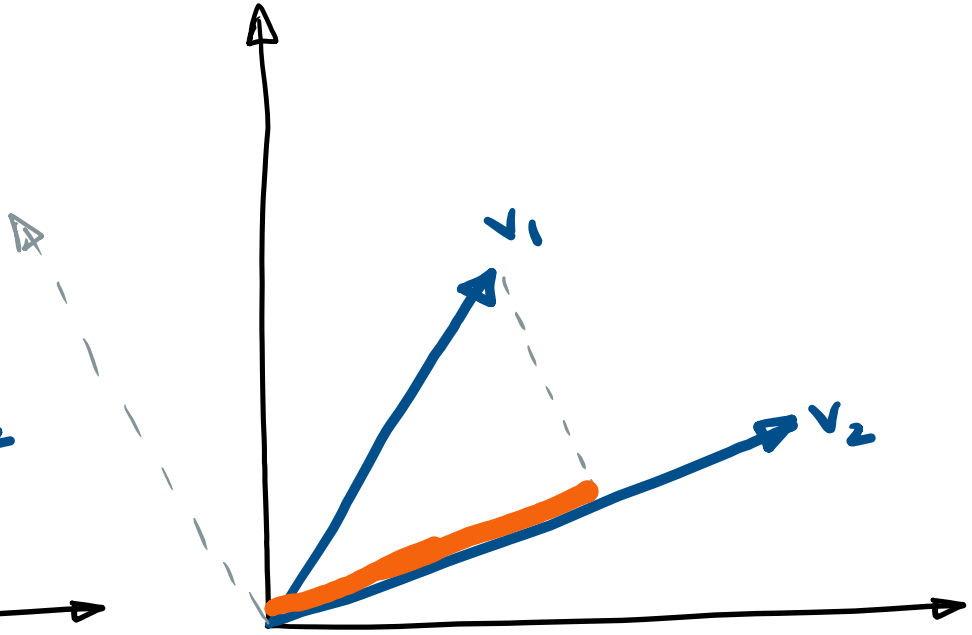
$$v_1 \cdot v_2$$

CROSS PRODUCT



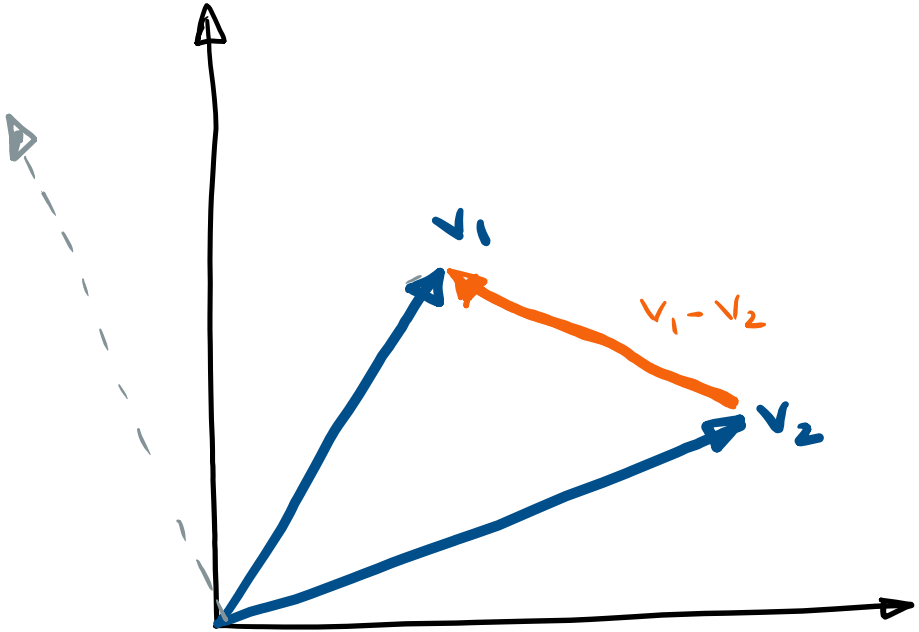
$v_1 \times v_2$

DOT PRODUCT

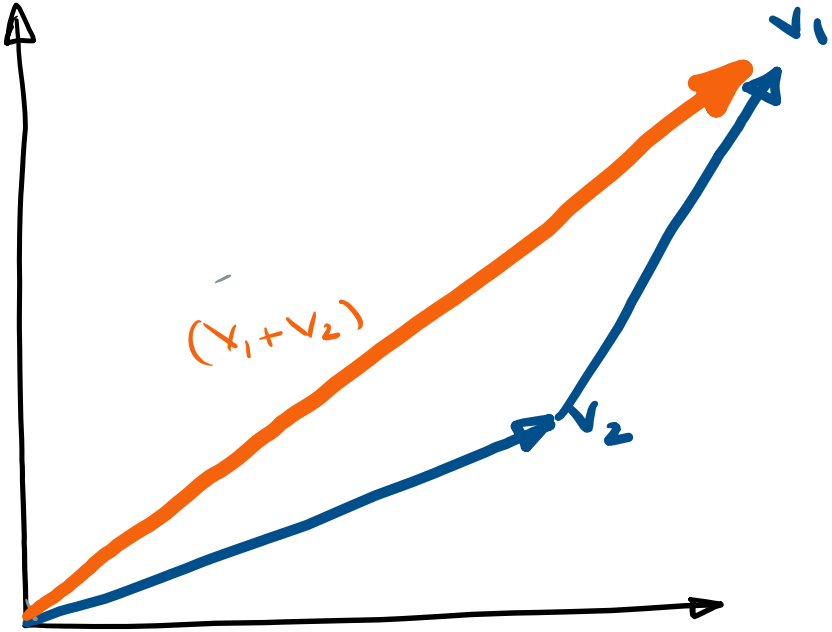


$v_1 \cdot v_2$

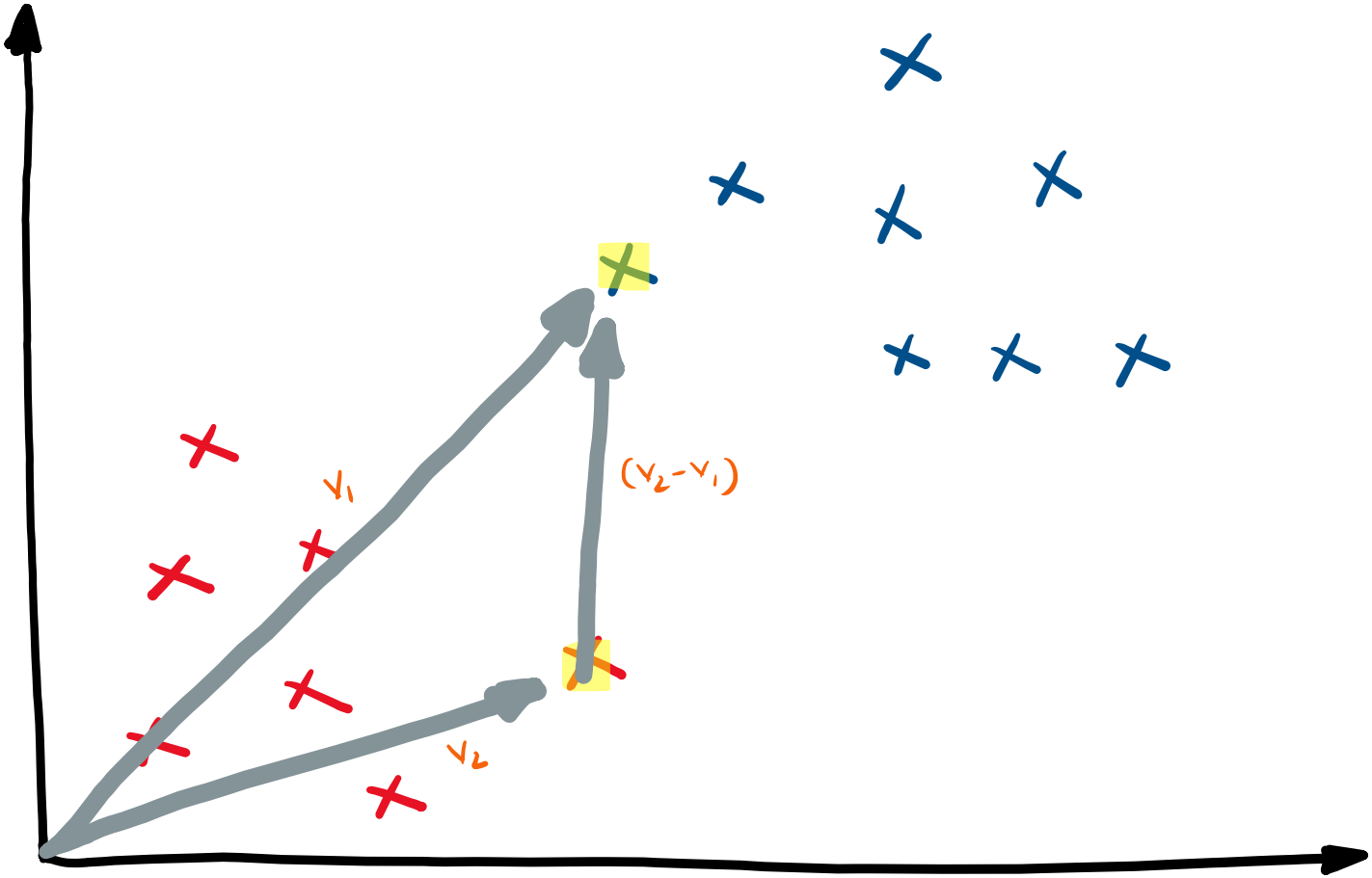
we can subtract v1 from v2



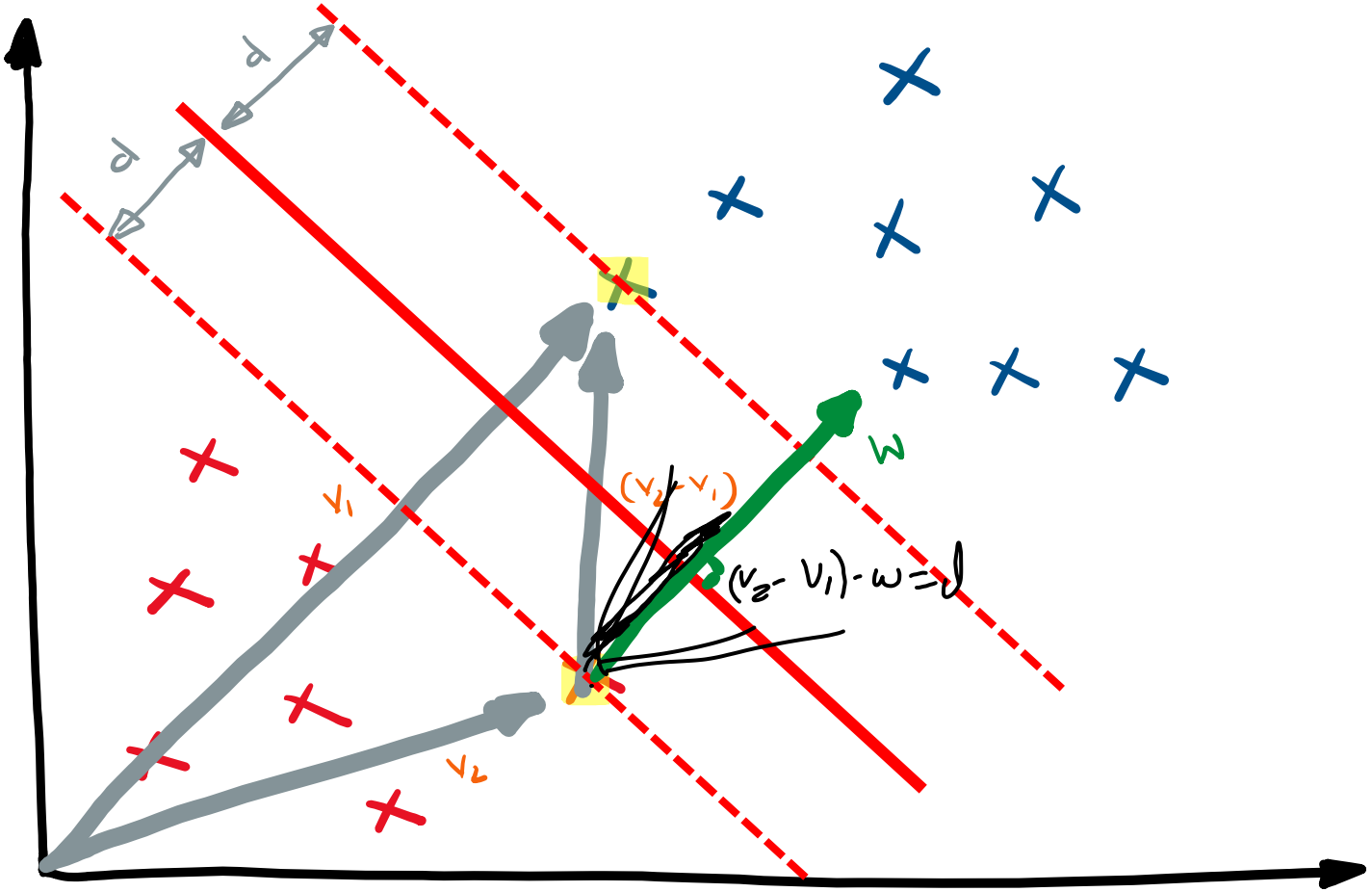
$(v_1 - v_2)$



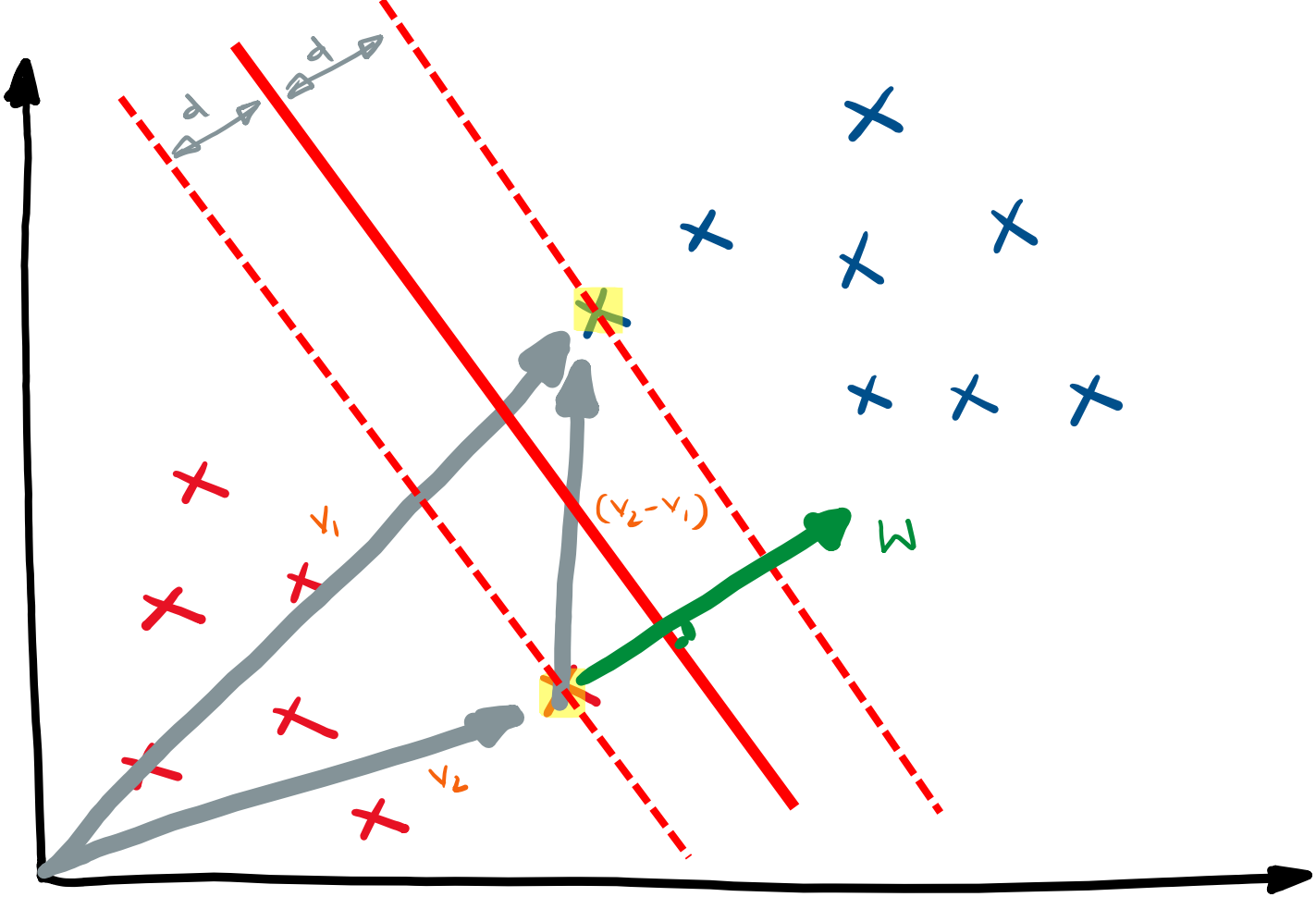
$(v_1 + v_2)$


v_1
$(v_2 - v_1)$
v_2

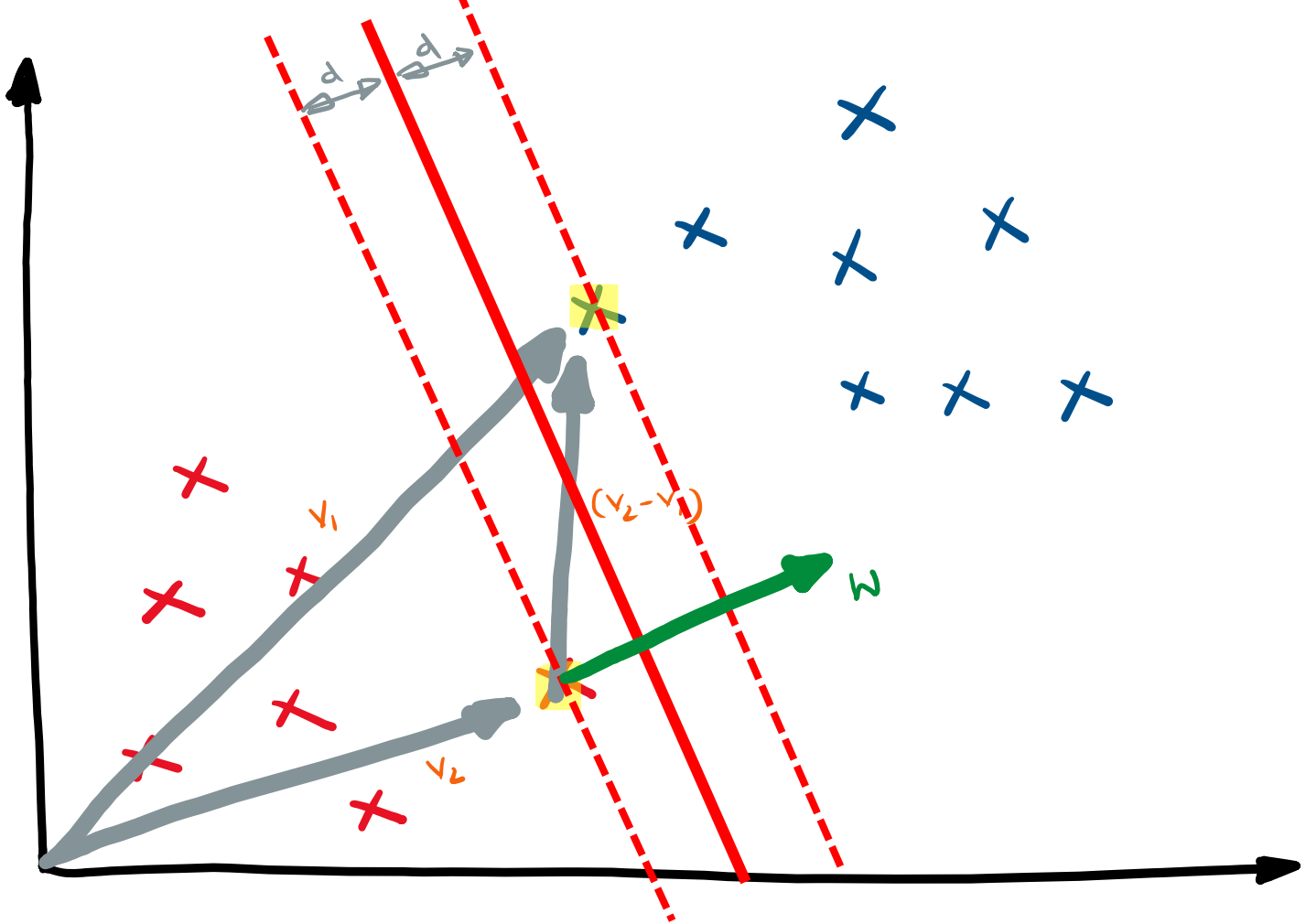
Case-1



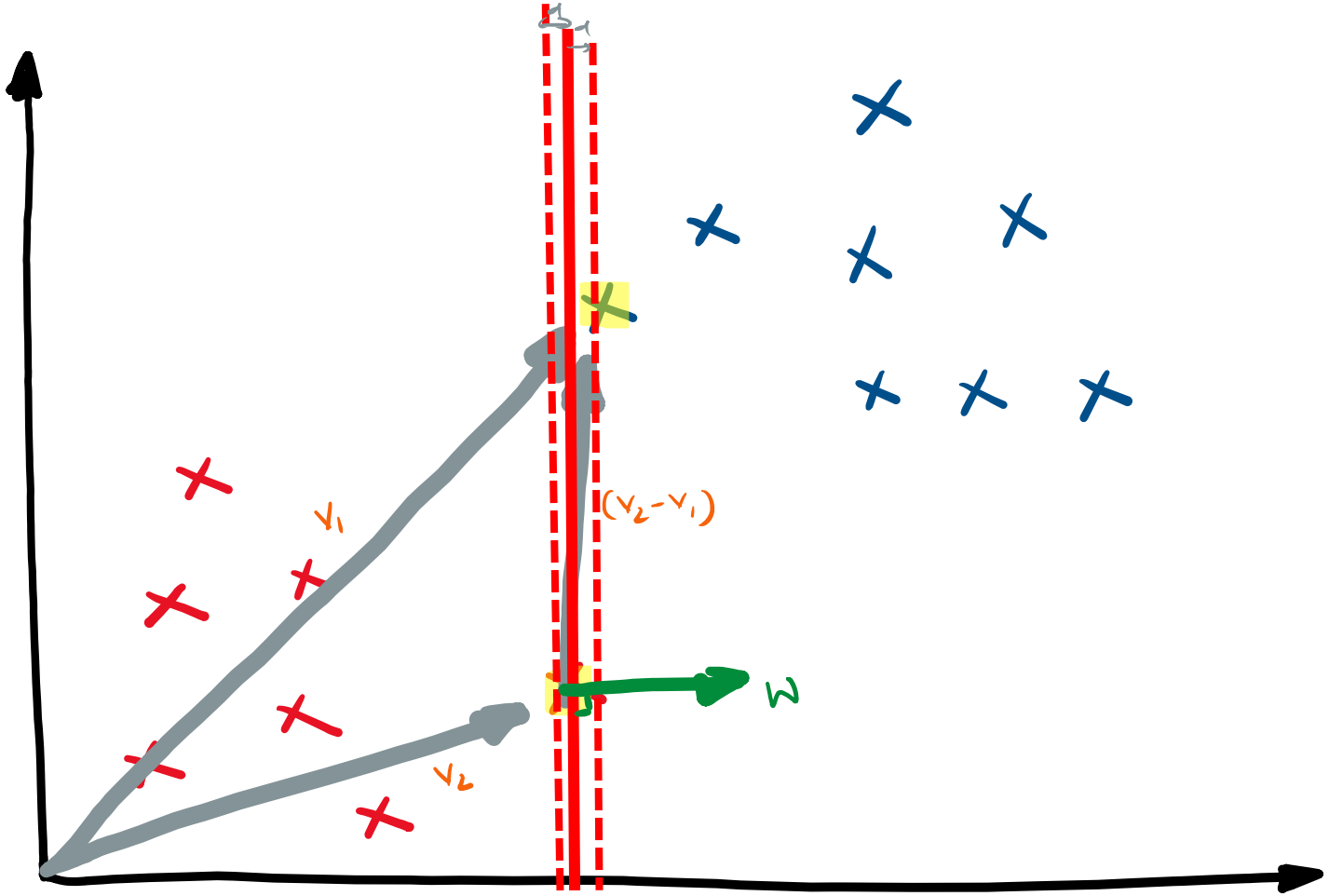
Case-2



Case-3

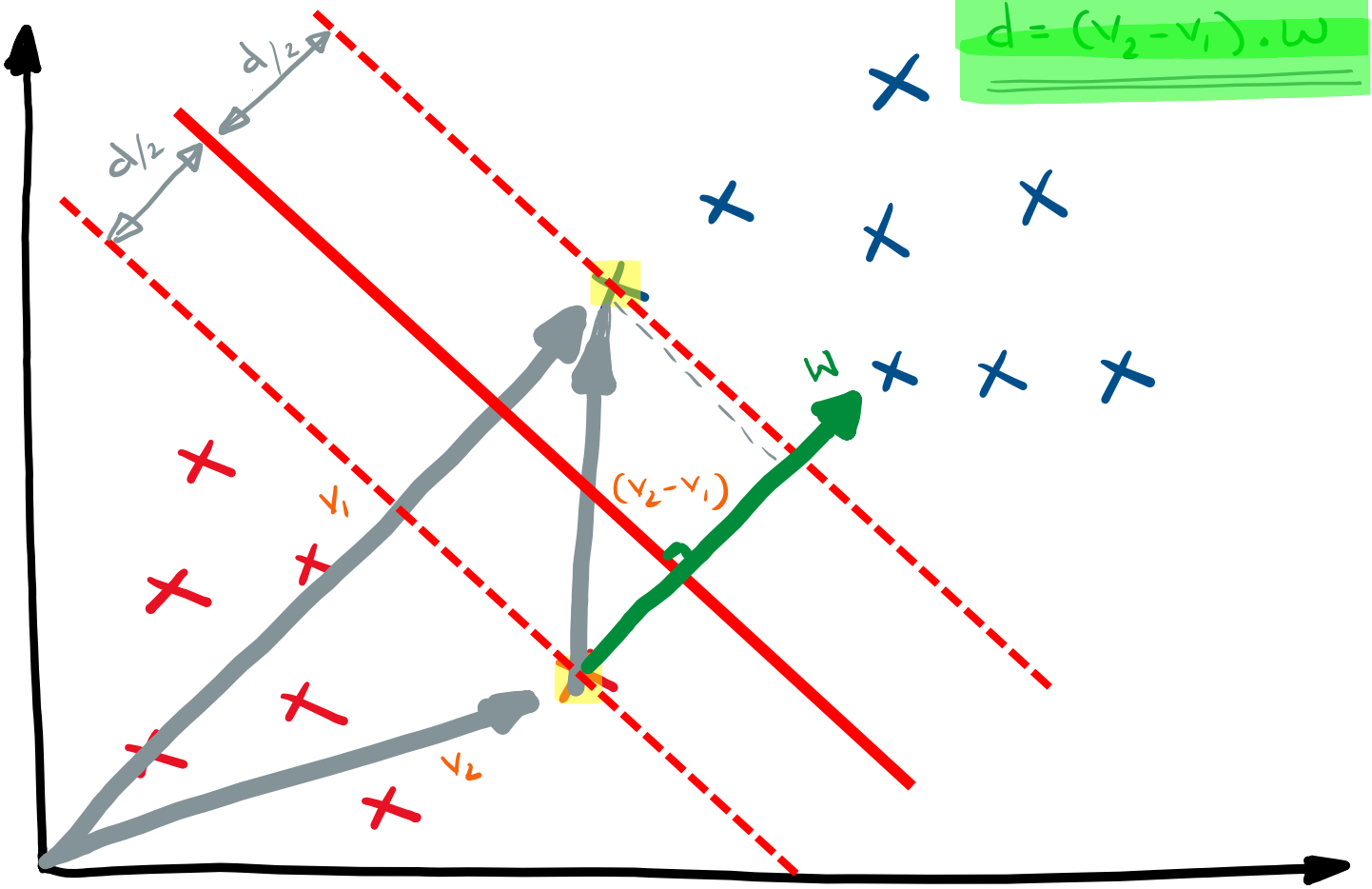


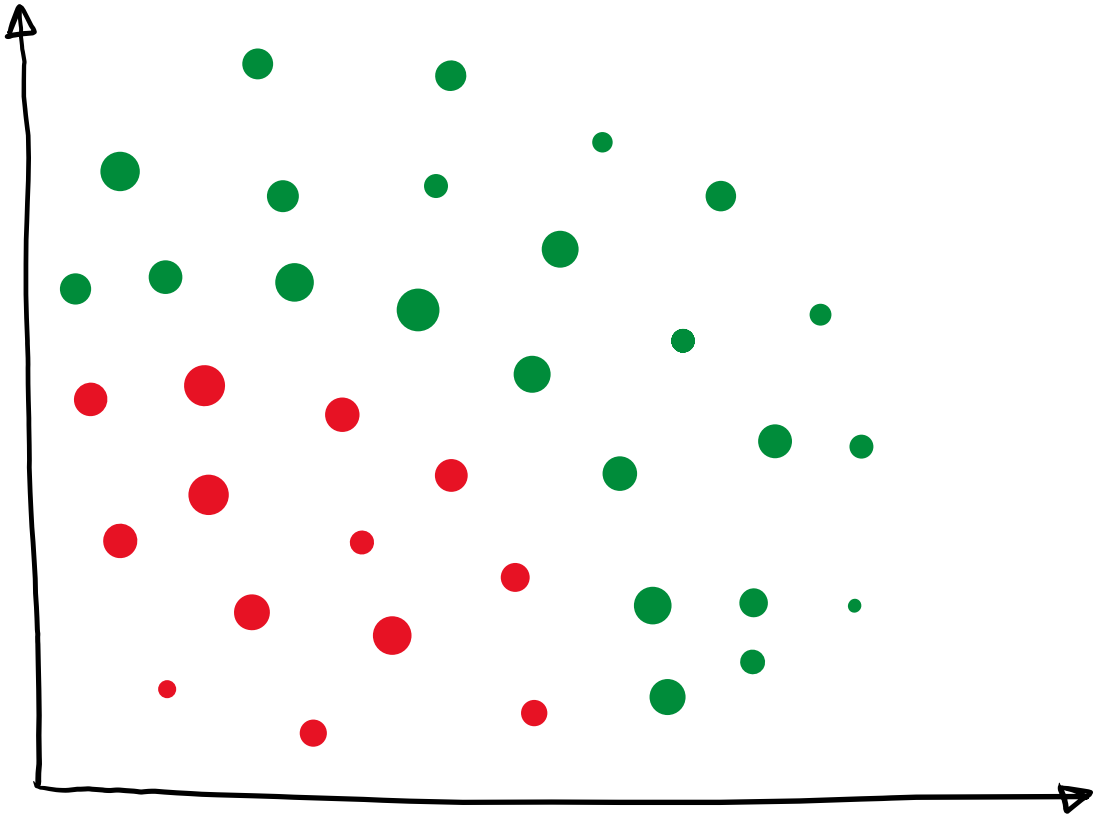
Case-4

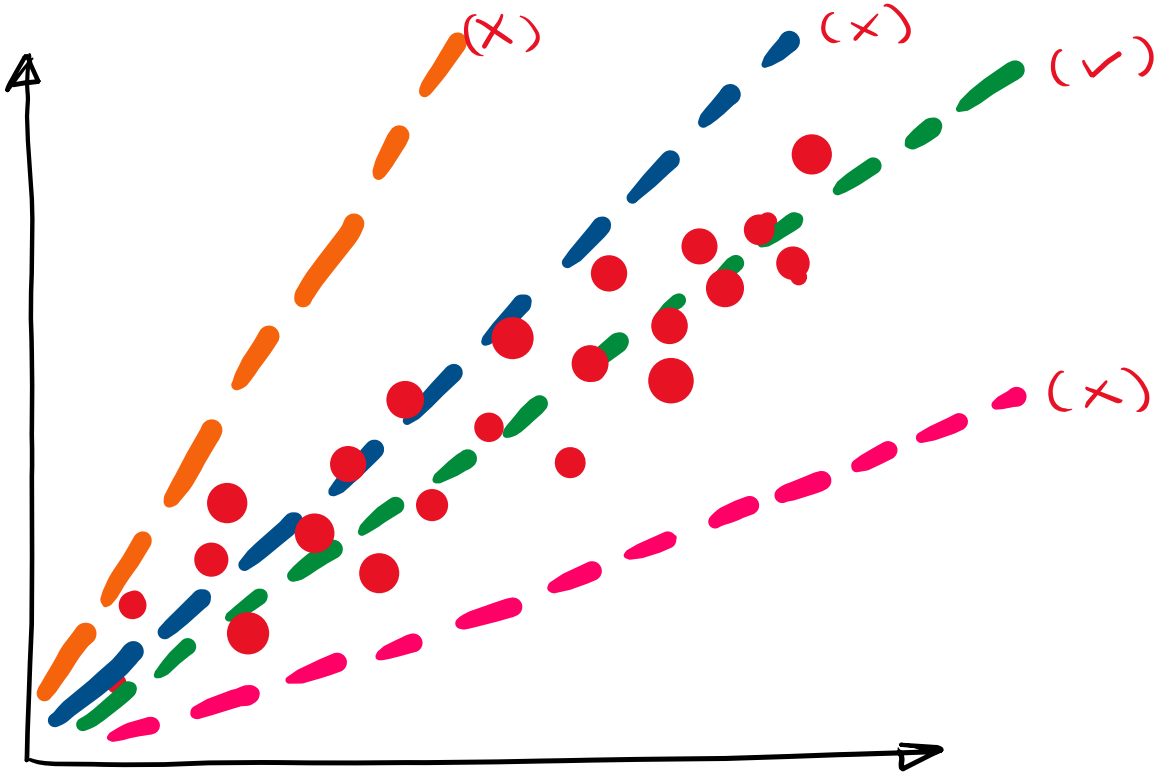


Case-1

$$d = (v_2 - v_1) \cdot w$$






(X)
(X)
(✓)
(X)

Advantages of SVM

1. More effective for high dimensional space
2. It is effective in cases where the number of dimensions is greater than the number of samples.
3. Can handles non-linear data efficiently by using the kernel trick (by adding one more dimension to data)
4. A small change to the data does not greatly affect the hyperplane and hence the SVM. So the SVM model is stable.

Disadvantage of SVM

1. Not suitable for large dataset. As training time is higher
2. SVM model is difficult to understand and interpret by human beings, unlike Decision Trees.
3. One must do feature scaling of variables before applying SVM
4. SVM doesn't directly provide probability estimates